

Homework 01

ECE 473/573, Fall 2026

Due Date: 09/06 (Sun.) by the end of the day (Chicago time)

(2 points) An important factor when making cloud computing decisions is its cost. Estimate the cost of renting the Linux server described in the syllabus: an x64 server with at least 4 CPU cores (vCPUs), 16GB of memory, and 100GB of storage. Consider both the pay-per-use and the subscription-based cloud services as follows for a workload of 10 hours per week for 17 weeks (or 4 months).

- Choose one well-known pay-per-use provider, such as AWS, Google Cloud, or Microsoft Azure. Estimate the total cost of an on-demand x64 Linux virtual machine that meets the requirements above. Avoid free-tier, promotional, reserved-instance, spot-instance, or commitment pricing.
- Choose one well-known subscription-based VPS (virtual private server) provider, such as those discussed in <https://joshtronic.com/2025/04/06/vps-showdown-april-2025-digitalocean-vs-linode-vs-upcloud-vs-vultr/>. Estimate the total cost of the x64 Linux plan to meet the same requirements.
- For both pricing models, you should report the instance type, the geographical location, vCPUs, memory size, storage type and size, network traffic allowance and overage charges, and urls to the web pages showing the instance or pricing.
- In a short conclusion, compare the two estimates. Which model is less expensive for the stated workload? What usage change could reverse your conclusion?

Here are a few hints.

- Pay-per-use compute is commonly priced per instance-hour for a fixed amount of CPU and memory combination.
- Persistent block storage allows you to save a machine without incurring computing cost. It is commonly priced per GB-month, and providers may offer different performance tiers.
- Inbound and outbound traffic are often priced differently. Verify the actual policy.

- While subscription-based VPS plans usually bundle CPU, memory, storage, and a monthly transfer allowance into one fixed price, additional services like backups, snapshots, public IPv4 addresses, or excess traffic may incur extra charges. Please make choices as you feel reasonable and state your assumptions.

Note that this homework does not require you to actually rent a server. Sometimes you could get more accurate pricing information by registering an account with the provider but for this homework you should NOT provide any payment information, e.g. credit card, during the process.